

# **Why Do Markets Matter?**

## **Specialization and Comparative Advantage**

**ECON 201: Principles of Microeconomics**  
**Lecture 4**

Div Bhagia

## What Does a Free Seat Cost?

A concert hall has unsold seats an hour before the show and lets students in free. The opportunity cost to the hall of admitting a student is:

- a) The regular ticket price.
- b) The hall's cost per seat for the evening.
- c) Close to zero, since the seat would otherwise be empty.
- d) Negative, since the student will buy snacks.

## The Sunk Cost Fallacy

The hall is paid for and the seat would otherwise sit empty. What does letting one more student in really cost?

- It is tempting to count the hall's spending on the evening, but that money is spent whether the student comes in or not.
- A **sunk cost** is a cost that has already been incurred and cannot be recovered.
- Letting sunk costs sway what you decide next is such a common mistake that it has its own name: the **sunk cost fallacy**.
  - For example, sitting through a movie you are not enjoying because you paid for the ticket.

## Where We Left Off

- Individuals make **choices** by weighing **costs** and **benefits**.
- The **economic cost** of an action is more than its direct cost:

$$\text{economic cost} = \text{direct cost} + \text{opportunity cost}$$

- **Opportunity cost**: the value of the next-best alternative you give up.
- So far, we have said that capitalism expanded our opportunities to specialize, by enlarging the role of both **firms** and **markets**.

**Today**: why specialization and trading make us better off.

## I, Pencil (Leonard Read, 1958)

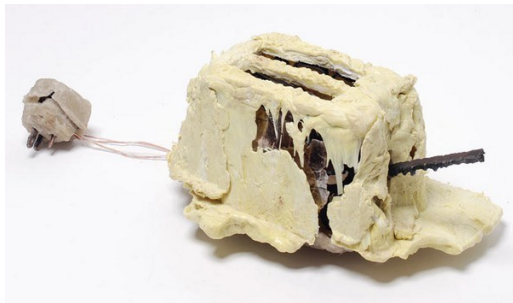
A pencil tells the story of how it was made:

- Wood from Oregon, graphite from Sri Lanka, brass from mined zinc and copper, an eraser from rapeseed oil.
- Thousands of workers involved, loggers, miners, sailors, none of whom set out to make a pencil.
- No single person knows how to make one from scratch.
- No one is in charge: markets coordinate all of them, Smith's **invisible hand** at work.



## The £4 Toaster

- In 2009, the designer Thomas Thwaites set out to build a toaster from scratch.
- He mined his own iron ore, smelted it, made plastic from potato starch, and drew his own copper wire.
- Nine months and about £1,187 later, his toaster worked for about five seconds before it began to melt.
- The store sold the same toaster for £3.94, about \$5.



## We All Specialize

**Specialization:** each person does just one or a few of the large number of tasks necessary to complete some project or reach some goal.

- **People** specialize: some weld, some teach, some design buildings.
- **Firms** specialize: often in a single step of making a single good.
- **Countries** specialize: gold and knitwear are Cambodia's largest exports; Singapore's are integrated circuits.

This seems obvious, but just two centuries ago, most families grew their own food and made much of their own clothing.

## Why Specialize?

- **Learning by doing:** we acquire skills as we produce things.
- **Differences in ability:** from skill, or from the soil, climate, and resources around us.
- **Economies of scale:** producing many units of a good often costs less per unit than producing a few.



## Meal Prep

Pablo and Lin are neighbors. Each spends about 8 hours on Sunday baking bread and making cheese for the week's meals. If they spent the whole day on just one food:

|       | Loaves of bread | Pounds of cheese |
|-------|-----------------|------------------|
| Pablo | 12              | 6                |
| Lin   | 2               | 4                |

They can also split the day: with half his time on each food, Pablo would make 6 loaves and 3 pounds of cheese.

**Think, pair, share:** Pablo can make more of **both** goods. Is there still a gain from trading with Lin? Think for a minute, then see what your neighbor thinks.

## Absolute Advantage

A person or a country has an **absolute advantage** in the production of a particular good if, given a set of available inputs, they can produce more of it than another person or country.

- Pablo has an absolute advantage in both goods: 12 loaves versus 2, and 6 pounds of cheese versus 4.
- So it may seem that Pablo has nothing to gain from trading with Lin.
- But that turns out not to be right. To see why, we need to look at each person's **opportunity cost** of producing each good, not at who makes more.

## Opportunity Costs

So what does each of them give up? Pablo's Sunday can produce 12 loaves or 6 pounds of cheese, so for Pablo each pound of cheese costs  $12 / 6 = 2$  loaves of bread.

| Opportunity cost of... | 1 pound of cheese    | 1 loaf of bread       |
|------------------------|----------------------|-----------------------|
| Pablo                  | $12 / 6 = 2$ loaves  | $6 / 12 = 0.5$ pounds |
| Lin                    | $2 / 4 = 0.5$ loaves | $4 / 2 = 2$ pounds    |

A pound of cheese costs Pablo four times as much bread as it costs Lin. The other way around, a loaf costs Lin four times as much cheese as it costs Pablo.

## Comparative Advantage

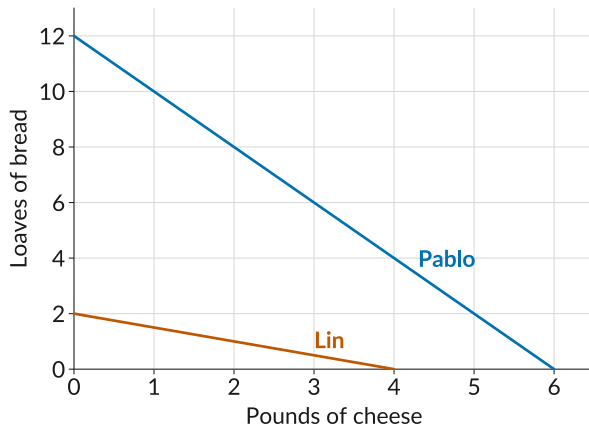
A person or a country has a **comparative advantage** in the production of a particular good if the cost to them of producing it, relative to the cost of another good, is lower than for another person or a country.

- Cost of a pound of cheese: 0.5 loaves for Lin, 2 loaves for Pablo.  
⇒ Lin has the comparative advantage in **cheese**.
- Cost of a loaf of bread: 0.5 pounds for Pablo, 2 pounds for Lin.  
⇒ Pablo has the comparative advantage in **bread**.
- Nobody can have a comparative advantage in everything: relatively cheaper at one food means relatively more expensive at the other.

## The Example in a Picture

- Each line shows every mix of bread and cheese one person can make.
- This is called their production possibilities frontier, or **PPF**.
- Pablo's line is **higher**; Lin's is **flatter**: cheese costs Lin less bread.

Height shows absolute advantage;  
slope shows comparative advantage.



## Self-Sufficiency

If they cannot trade, each eats exactly what they make. Suppose each of them spends **75% of their time on bread and 25% on cheese.**

|              | Loaves of bread       | Pounds of cheese      |
|--------------|-----------------------|-----------------------|
| Pablo        | $0.75 \times 12 = 9$  | $0.25 \times 6 = 1.5$ |
| Lin          | $0.75 \times 2 = 1.5$ | $0.25 \times 4 = 1$   |
| <b>Total</b> | <b>10.5</b>           | <b>2.5</b>            |

Under **self-sufficiency**, each person consumes exactly what they produce.

## Specializing by Comparative Advantage

Now suppose **each of them makes only the food in which they have the comparative advantage**: so Pablo makes bread, and Lin makes cheese.

|                      | Loaves of bread | Pounds of cheese |
|----------------------|-----------------|------------------|
| On their own (total) | 10.5            | 2.5              |
| Specialized          | <b>12</b>       | <b>4</b>         |

The two of them now produce **1.5 more loaves and 1.5 more pounds of cheese** than before, with the same skills and the same hours. Total output went up just from rearranging who does what.

## Specialization Without Trade

Total output is higher, but look at what each of them ends up with.

|       |        | On their own | Specialized |
|-------|--------|--------------|-------------|
| Pablo | Bread  | 9            | 12          |
|       | Cheese | 1.5          | 0           |
| Lin   | Bread  | 1.5          | 0           |
|       | Cheese | 1            | 4           |

Pablo has 12 loaves and no cheese. Lin has 4 pounds of cheese and no bread. Specializing is only worthwhile if they can **trade**.



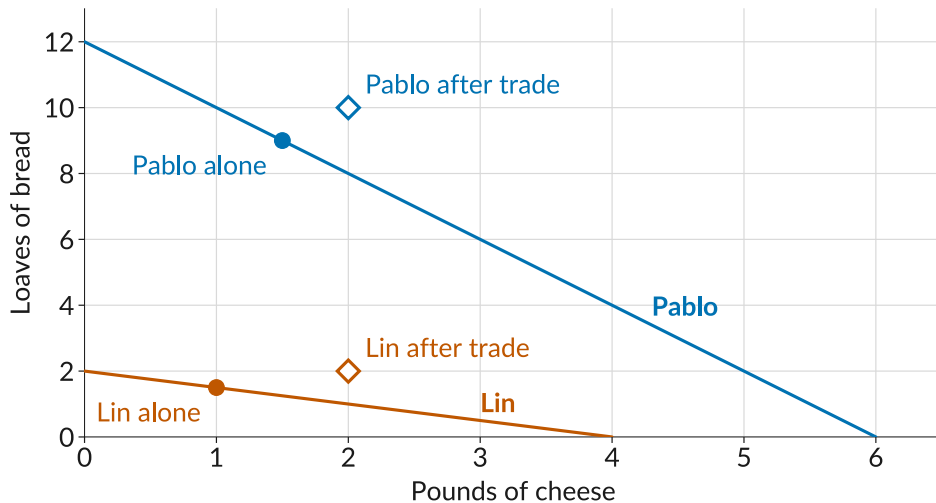
## Specialization and Trade

Now suppose they agree on a deal: **Lin sells Pablo 2 pounds of cheese for 2 loaves of bread.**

|              |        | On their own | Special-<br>ized | Trades         | Consumes  |
|--------------|--------|--------------|------------------|----------------|-----------|
| Pablo        | Bread  | 9            | 12               | <b>sells 2</b> | <b>10</b> |
|              | Cheese | 1.5          | 0                | <b>buys 2</b>  | <b>2</b>  |
| Lin          | Bread  | 1.5          | 0                | <b>buys 2</b>  | <b>2</b>  |
|              | Cheese | 1            | 4                | <b>sells 2</b> | <b>2</b>  |
| <b>Total</b> | Bread  | 10.5         | 12               |                | <b>12</b> |
|              | Cheese | 2.5          | 4                |                | <b>4</b>  |

Compare the first column with the last: both of them now consume more of both goods.

## Trade in the Picture



## Why Did That Work?

Pablo bought 2 pounds of cheese from Lin, even though Pablo could have made them in less time. Why does that make sense?

- Every hour Pablo spends on cheese costs a lot of bread. An hour of Lin's costs only a little.
- What matters is not who is better at making cheese, but whose cheese costs less **in bread**. That is Lin.

Someone can be worse at producing everything and still have a comparative advantage in something.

## Sharing the Gains from Trade

- Their deal, 2 pounds of cheese for 2 loaves, comes down to a **price**: 1 loaf per pound of cheese.
- Neither will trade at a price worse than making the food on their own:
  - Lin sells only if the price is above 0.5 loaves per pound, the cost of making cheese.
  - Pablo buys only if the price is below 2 loaves per pound, the cost of making cheese at home.
- Any price between 0.5 and 2 loaves per pound makes both of them better off. Prices near 0.5 favor Pablo, and prices near 2 favor Lin.

## Questions

- Suppose Taylor Swift types faster than any assistant. Should she answer her own fan mail?
- You are better than your roommate at both cooking and cleaning. Should you do both?
- Should the United States import goods that it could produce more cheaply at home?
- Are there any downsides to specializing?

## What to Do Next

- Before next class: read section **2.3**, go over the **slides** and the **worksheet**, and work the **practice problems**.
- While driving or at the gym, listen to **An Economics Lesson from a Talking Pencil**, from Freakonomics Radio.
- No class Monday (Labor Day). **Quiz 2 is Wednesday, September 9** (Lectures 3 and 4).
- Next week we go inside the firm: what it costs to make things, and how a firm chooses its price.